

JEE MAIN REVISION MASTER: IUPAC NOMENCLATURES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 5 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	4 PYQs	Applied Topics
Standard	Basics Of Organic Chemistry — Core Principles & Governing Laws	1 PYQs	Core Principle
Standard	Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	$\text{HC} \equiv \text{C}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2, \quad \text{C}_{20}\text{H}_{42}, \quad \text{C}_n\text{H}_{2n+2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basics Of Organic Chemistry — Applied Cases, Reactions & Traps (Moderate Yield — 4 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound with molecular formula $\text{C}_{20}\text{H}_{42}$ is..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "The IUPAC name of the given organic compound is $\text{HC} \equiv \text{C}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "IUPAC name of the compound is..." [Ans: N/A]

3.2 Basics Of Organic Chemistry — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Basics Of Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Basics Of Organic Chemistry PYQs.

Representative Source PYQs:

- [2021 Shift] Q#1 (MCQ): "The correct IUPAC name of..." [Ans: N/A]

3.3 Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound with molecular formula $C_{(20)}H_{(42)}$ is..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "The IUPAC name of the given organic compound is $HC \equiv C-CH=CH-CH=CH_{(2)}...$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The compound with molecular formula $C_{(20)}H_{(42)}$ is

- (A) Decane
- (B) Dodecane
- (C) Eicosane [Correct]
- (D) Hicosane

Correct Answer: C

Solution Analysis: The molecular formula is $C_{(20)}H_{(42)}$. For alkanes, the general formula is $C_{(n)}H_{(2n+2)}$. Here, $n=20$. So, $H=2(20)+2=42$. Hence the compound is an alkane with 20 carbon atoms. The alkane with 20 carbon atoms is called Eicosane. So, the correct option is Option C: Eicosane.